



GDCAU

Green Deal for Central Asian Universities

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What is Regeneration?

1. Why Regeneration Matters Today

Across Central Asia and beyond, ecosystems and communities are facing rapid and interlinked ecosystems pressures: shrinking glaciers, water stress, soil degradation, desertification, and biodiversity decline that have societal impact. While sustainability remains essential, many researchers argue that **reducing harm alone is not enough** when systems are already strained.

That is where **regeneration** comes in. Regeneration is a complementary lens that focuses on **repairing, renewing, and strengthening** the capacities of living systems—ecological and social—so they can adapt to change and support flourishing futures. It invites students, communities, and institutions to engage directly in restoring the conditions that make life possible.

Recent scholarship in ecology, biology, land management, and the social sciences all point toward the same idea: our future depends on learning how to work with living systems, not just manage them.

The European Green Deal reflects this shift as well: its policies increasingly support not only sustainability but also **restorative and regenerative approaches**, such as ecosystem restoration, circular economy design, soil health, and nature-based solutions. Regeneration aligns with the EU's long-term goal of creating resilient, thriving socio-ecological systems—and it provides students with the mindset and skills needed to participate in that transition.

2. What Regeneration Is

Regeneration is the process of restoring, renewing, and enhancing the vitality of systems (ecological, social, and cultural) so they become healthier, more resilient, and more capable of supporting the ongoing conditions for renewal and flourishing over time.

Regeneration is not a rejection of sustainability but an **extension** of it. Where sustainability asks us to reduce harm, regeneration asks how we can **improve the system**—its soil, water, biodiversity, relationships, governance, culture, and community well-being.

Key qualities of regenerative approaches

- **Life-enhancing:** increases ecological health and social well-being.
- **Adaptive:** welcomes change and variation instead of resisting them.
- **Relational:** recognizes the links between water, soil, climate, communities, and livelihoods.
- **Place-based:** shaped by local landscapes, cultures, and histories.
- **Future-oriented:** builds capacity for long-term resilience and thriving.

Grounding in contemporary research (see “*Further Reading*” section below for details)

- **Ecology:** Restoration and rewilding studies show that degraded ecosystems, from wetlands to grasslands, can regenerate through improved management and reduced pressure.
- **Regenerative agriculture:** Demonstrates how soil can regain fertility, carbon storage, and biodiversity when farming practices align with ecological processes (Schreefel et al., 2020).
- **Regenerative design:** Emphasizes positive cycles (diversity, mutualism, feedback awareness) that strengthen system health (Buckton, Fazey).
- **Social Science:** Thinkers like Latour, Bennett, Ingold, and Tsing show how humans and nonhumans co-create the world through entanglement, relationality, and material agency.
- **Systems thinking:** Modern regeneration approaches emerge from systems science, drawing on concepts such as feedback loops, adaptive cycles, thresholds, and cross-scale interactions (Holling; Meadows; Berkes; and Folke). These frameworks provide the analytical foundations for understanding how living systems change—and how regeneration strengthens their capacity to thrive.

3. Sustainability, Resilience, and Regeneration — Three Connected Lenses

Rather than opposing them, it is helpful to see these as **nested approaches**.

Sustainability

- Prevents further degradation
- Encourages responsible use of resources
- Sets targets and standards for long-term planning

(Adaptive) Resilience

- Helps systems absorb shocks and reorganize
- Focuses on adaptation and learning under uncertainty
- Draws from socio-ecological systems (SES) thinking

Regeneration

- Builds on both
- Seeks to **increase the health** of systems
- Brings together ecological restoration, community vitality, and cultural meaning

4. Regeneration in Practice: Models and Possibilities

Regeneration is not only about healing ecosystems—it is also about strengthening the **social relationships, cultural practices, and community capacities** that allow ecological recovery to last. Around the world, successful regeneration projects show that **ecological vitality and social well-being rise together**, because people and landscapes are part of the same system.

International Example: Vattenrike, Sweden

- Once dismissed as degraded wetlands, Vattenrike became a living model of socio-ecological renewal after local teachers, farmers, fishers, and policymakers worked together to revalue the wetland's cultural and ecological importance.
- Ecological monitoring restored bird habitats and water quality, **while new partnerships created jobs, education programs, cultural events, and eco-tourism.**
- Researchers describe Vattenrike as an example of **adaptive co-management**, where community engagement and ecological science support one another to create long-term resilience.

Global Rewilding Examples

Rewilding is an approach to ecological restoration that allows natural processes to return—often by reducing human pressures and, when appropriate, reintroducing key species that shape ecosystem dynamics.

Iberian Peninsula (Portugal & Spain)

- Rewilding projects here restored keystone species like Iberian horses and vultures, which helped reduce wildfire risk by managing vegetation naturally.
- Local shepherds, foresters, and youth groups participated in new land stewardship roles, creating **hybrid ecological–social economies** (eco-tourism, grazing cooperatives, habitat management jobs).

Carpathian Mountains (Romania)

- Forest rewilding improved biodiversity, but just as importantly, it involved **local villages in wildlife monitoring, cultural festivals, and sustainable forestry cooperatives**.
- Research shows that community participation increased trust, reduced conflict, and strengthened social cohesion.

Highlands & Community Rewilding (Scotland)

- Projects like *Trees for Life* and community-led land trusts have restored native woodlands and peatlands while supporting the regeneration of Highland villages through local governance, cultural revival, and new livelihood opportunities in conservation, eco-tourism, and land stewardship.

African Landscape Rewilding (Kenya, Zimbabwe, South Africa)

- Many African rewilding programs combine ecological restoration with **community conservancies**, where local residents co-manage land and receive direct benefits from conservation (jobs, revenue-sharing, education programs).
- These initiatives show how regeneration succeeds when ecosystems thrive *and* communities have a stake in renewal.

Central Asian Analogues

Aral Sea Basin

- Tree planting and dune stabilization projects not only reduce dust storms and restore soil—they also involve **local schools, women’s cooperatives, youth groups, and farmers**, building social networks of care around land restoration.

Community-Based Pasture Management in Kyrgyzstan

- Over the past decade, pasture committees (jaiyt committees / ЖАЙЫТ КОМИТЕТТЕРИ) have re-established **traditional knowledge**, rotational grazing practices, and democratic decision-making.
- These efforts improve soil and vegetation health while increasing **community cohesion, shared responsibility, and conflict resolution capacity**.

Glacier Monitoring & Watershed Protection

- Student-led glacier monitoring and citizen-science river projects link ecological data with community priorities such as drinking water security, irrigation reliability, and climate adaptation.
- These projects strengthen both **ecological insight** and **local adaptive capacity**.

Urban Greening & Ecological Corridors

- Cities like Bishkek, Almaty, Dushanbe, and Tashkent are planting street trees, restoring riverbanks, and creating green corridors that serve ecological functions *and* provide safer, healthier public spaces.
- Urban regeneration projects often involve residents in planting days, environmental education, and neighborhood maintenance—supporting both biodiversity and **community identity**.

Why These Examples Matter

Across these cases, regeneration works because it strengthens **both**:

- **ecological systems** (soil, water, biodiversity, habitat)
- **social systems** (trust, cooperation, local knowledge, cultural meaning, livelihoods)

Research in socio-ecological resilience shows that **ecosystems recover more fully and remain healthy longer when communities are engaged, empowered, and connected**.

5. What Regeneration Means for You

This guide prepares you to approach your studies and your future career with a **regenerative mindset**.

As a student, regeneration invites you to:

- Notice how land, water, climate, and social life connect
- Develop systems thinking skills
- Recognize feedback loops that shape outcomes
- Learn from community experts, elders, and local knowledge-holders
- Understand the ecological signals of your region
- Try small regenerative experiments:
 - biodiversity patches
 - soil observation plots
 - water flow mapping
 - campus circularity projects
- Reflect, iterate, and adapt

Regeneration becomes not just an idea, but a **practice of noticing, caring, and collaborating with the world around you**.

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Bremen, Germany
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Further Reading

Socio-Ecological Systems

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